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DEPARTMENT OF COMPUTER APPLICATIONS - MCA

Mini - Project Proposal

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BioDiversity Monitoring Dashboard

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Abstract

Urban green spaces play a crucial role in maintaining biodiversity and promoting ecological balance within cities, which are often rapidly expanding due to urbanization and industrialization. These green spaces are vital not only for the well-being of residents but also for sustaining various ecosystems and providing essential services such as air purification, temperature regulation, and recreational spaces. However, many urban environments, particularly in India, are facing significant challenges related to biodiversity loss. This decline is primarily caused by a combination of unmonitored species interactions, the unchecked spread of invasive species, and insufficient public engagement in conservation efforts. These issues threaten the ecological health of urban ecosystems, which may otherwise provide critical refuges for local flora and fauna.

In response to these challenges, this project proposes the development of a **Biodiversity Monitoring Dashboard** designed specifically for urban green spaces. This platform integrates **citizen science data** with advanced ecological analytics, enabling residents and local communities to actively participate in the monitoring and conservation of urban biodiversity. The system empowers users to upload geotagged species sightings, which are crucial for tracking species distribution and abundance over time. Additionally, it computes key biodiversity indices, such as the **Shannon Diversity Index**, which provides a measure of species diversity and ecosystem health.

One of the unique features of the proposed solution is its ability to assess the risk posed by **invasive species** using spatial analysis. By evaluating the spread of non-native species within urban green spaces, the system can help identify areas at high risk of invasion, which is critical for early intervention and preventing further ecological disruption. The platform not only serves as a tool for data collection but also acts as a decision-support system for urban planners, conservationists, and policymakers to design informed, data-driven conservation strategies.

The overarching goal of this project is to create a user-friendly and accessible platform that encourages **community-driven biodiversity monitoring**. By bridging the gap between raw citizen science data collection and sophisticated ecological modeling, the system fosters local involvement and encourages

Introduction

Urbanization has dramatically altered natural landscapes, leading to widespread **habitat fragmentation** and a sharp decline in biodiversity. As cities expand, vast areas of **natural ecosystems**—such as forests, wetlands, and grasslands—are replaced with impervious surfaces, such as roads, buildings, and industrial zones, which disrupt the continuity of wildlife habitats. This fragmentation not only isolates species but also diminishes the quality of their habitats, making them more vulnerable to external threats.

Urban green spaces, such as parks, wetlands, and institutional campuses, have become crucial sanctuaries for native flora and fauna. These areas serve as the last bastions of biodiversity in otherwise concrete-dominated environments, providing habitats for a wide variety of species and helping to maintain ecological processes within urban landscapes.

Despite their ecological importance, urban green spaces are often poorly monitored, with limited efforts made to track species populations, monitor ecosystem health, and assess environmental stressors. Without systematic, continuous tracking, the loss of biodiversity in these spaces often goes unnoticed until it reaches critical levels. This delay in detection can lead to the spread of **invasive species**, the decline of native species, and the erosion of ecosystem services that these areas provide.

The phenomenon of the "**extinction of experience**," in which urban residents become disconnected from nature, further exacerbates the problem, making it more difficult to generate public awareness or support for conservation efforts. Without direct engagement with the natural world, citizens may fail to recognize the intrinsic value of green spaces, undermining efforts to protect them.

The ecological integrity of urban "islands" of biodiversity is further threatened by the accelerating pace of **city expansion**. As cities grow, development projects often prioritize **grey infrastructure**—such as roads, buildings, and utilities—over the preservation of **green ecosystems**. The resulting habitat loss, combined with the degradation of remaining green patches, creates an environment where urban ecosystems are increasingly susceptible to **edge effects** and environmental stressors. These effects, which include changes in microclimate, increased exposure to pollutants, and reduced habitat connectivity, make it even harder for native species to thrive in urban settings.

Objectives and Scope of the Project.

Objectives

Objectives

- 1. Develop a web-based platform for citizen biodiversity reporting:**
The platform will allow urban residents to report geotagged sightings of species in local green spaces, promoting **citizen science** for urban biodiversity monitoring.
- 2. Compute and visualize ecosystem health metrics:**
The system will calculate and display key ecological indices, such as the **Shannon Diversity Index** and **species richness**, to assess and visualize the health of urban ecosystems.
- 3. Analyze invasive species spread and assess ecological risks:**
The platform will track the spread of invasive species in urban areas through **spatial analysis**, allowing early detection and risk assessment for ecological threats.
- 4. Provide conservation recommendations:**
Based on the data collected, the system will offer actionable insights and recommendations for **biodiversity conservation and restoration**, helping to guide local conservation strategies.
- 5. Encourage public engagement and transparency:**
The platform will feature a transparent interface showing the impact of individual contributions, fostering public involvement and environmental stewardship.

Scope of the Project

- Focus on Urban Green Spaces in India:**
The project will target **urban green spaces** in Indian cities, focusing primarily on **parks, wetlands, and institutional campuses**.
- Biodiversity Data Collection:**
The system will collect and analyze sightings of **flora and fauna**, including **birds, plants, insects, and small mammals**, to assess biodiversity in urban ecosystems.
- Data Processing for Analysis:**
A minimum of **200 sightings** will be processed for each green space to provide statistically significant data for biodiversity index computation and risk assessment.

Literature review

(Summary of Existing Research).

Several studies emphasize the importance of biodiversity monitoring in urban ecosystems to ensure ecological sustainability. Citizen science platforms such as eBird and iNaturalist have demonstrated success in collecting large-scale biodiversity data. Ecological indices, including the Shannon and Simpson Diversity Indices, are widely used to quantify species diversity and ecosystem stability.

Recent academic discourse highlights the shift from traditional "top-down" monitoring to "bottom-up" citizen-led initiatives. Research indicates that while professional ecological surveys provide high precision, they often lack the temporal frequency needed to track rapid urban changes. In contrast, digital dashboards that leverage crowd-sourced data provide a continuous stream of information, allowing for a more dynamic understanding of urban phenology. Studies in the *Journal of Urban Ecology* suggest that the integration of GIS (Geographic Information Systems) with citizen science significantly increases the detection rate of rare species and the early reporting of environmental degradation in fragmented landscapes.

Furthermore, the role of data visualization in environmental policy cannot be overstated. Literature concerning "Information Design for Conservation" demonstrates that complex ecological data, when presented through interactive dashboards, is more likely to influence local government decision-making. By transforming abstract diversity scores into visual heatmaps and trend lines, technical platforms bridge the communication gap between scientists and urban planners. This allows for more targeted interventions, such as the creation of "green corridors" to connect isolated habitat patches.

Methodology or Approach Planned.

1. Data Collection

Use the Urban Green Space Dataset obtained from Kaggle as the base dataset.

Allow users to upload species sightings with photographs, geolocation, and timestamps.

2. System Architecture

Frontend (React JS):

User interface for uploading species sightings with image previews.

Interactive charts for displaying biodiversity indices and trends.

Alerts and notifications for invasive species threats.

Backend (SQL Database):

Tables for species sightings, species inventory, and habitat zones.

Spatial SQL queries for geotagged biodiversity analysis.

Aggregate functions for computing ecological indices such as the Shannon Diversity Index using:

$$H = -\sum (p \times \ln(p))$$

3. Analysis Components

Monitoring: Calculation of species richness and diversity scores with temporal comparison.

Invasion Assessment: Spatial modeling of invasive species spread and habitat encroachment.

Interpretation: Correlation analysis between native species decline and invasive species growth, along with conservation recommendations.

4. Evaluation Metrics

A minimum of 200 validated biodiversity sightings.

Biodiversity index stability greater than 85%.

Validation of invasion risk predictions using historical data patterns.

Expected Outcomes/Designs of Outcomes

- A fully functional biodiversity monitoring dashboard for urban green spaces: The primary deliverable is a responsive web application that serves as a centralized hub for ecological data. This includes a secure user authentication system, a seamless media upload pipeline, and a dynamic administrative interface for data moderation.
- Visualization of ecosystem health indicators such as species diversity and richness: The system will generate high-fidelity visual representations of complex biological data. Using libraries like D3.js or Chart.js, the dashboard will provide real-time updates to the Shannon Diversity Index () and species richness counts. Users will be able to filter these visualizations by specific date ranges, habitat types, or taxonomic groups.
- Early detection and risk categorization of invasive species: One of the most critical outcomes is an automated spatial alert system. By comparing new sightings against a known database of invasive flora and fauna, the dashboard will categorize threats based on their proximity to sensitive native habitats. This provides a "heat map" of potential ecological encroachment, allowing for rapid intervention by local authorities.
- Data-driven insights to support urban biodiversity conservation and restoration efforts: Beyond raw data, the platform will generate summary reports that correlate environmental factors with species presence. For instance, the system might highlight a correlation between the presence of specific native pollinators and the health of local flora. These insights will be formatted to assist urban planners in designing "Green Infrastructure" that supports local wildlife.
- A scalable framework suitable for future expansion to additional cities or datasets: The backend architecture is designed with modularity in mind. By utilizing standardized SQL schemas and a flexible React frontend, the platform can be easily replicated for other metropolitan areas in India. Future integrations could include API hooks for weather data or air quality sensors to provide a holistic view of the urban environment.
- A comprehensive project report documenting the correlation between urban density and biodiversity health within the studied green spaces: A final academic outcome will be a detailed analysis of how urban stressors—such as proximity to high-traffic roads or residential density—affect the biodiversity indices calculated by the dashboard. This report will serve as a scientific validation of the project's methodology and findings.